$with(LinearAlgebra):$

$A := Matrix([[1, x0, x0^2, x0^3], [1, x1, x1^2, x1^3], [1, x2, x2^2, x2^3], [1, x3, x3^2, x3^3]]);$

$$A := \begin{bmatrix} 1 & x0 & x0^2 & x0^3 \\ 1 & x1 & x1^2 & x1^3 \\ 1 & x2 & x2^2 & x2^3 \\ 1 & x3 & x3^2 & x3^3 \end{bmatrix} \tag{1}$$

$d := Determinant(A);$

$$d := x0^3\, x1^2\, x2 - x0^3\, x1^2\, x3 - x0^3\, x1\, x2^2 + x0^3\, x1\, x3^2 + x0^3\, x2^2\, x3 - x0^3\, x2\, x3^2 - x0^2\, x1^3\, x2 + x0^2\, x1^3\, x3 + x0^2\, x1\, x2^3 - x0^2\, x1\, x3^3 - x0^2\, x2^3\, x3 + x0^2\, x2\, x3^3 + x0\, x1^3\, x2^2 - x0\, x1^3\, x3^2 - x0\, x1^2\, x2^3 + x0\, x1^2\, x3^3 + x0\, x2^3\, x3^2 - x0\, x2^2\, x3^3 - x1^3\, x2^2\, x3 + x1^3\, x2\, x3^2 + x1^2\, x2^3\, x3 - x1^2\, x2\, x3^3 - x1\, x2^3\, x3^2 + x1\, x2^2\, x3^3 \tag{2}$$

$factor(d);$

$$(x2 - x3)\,(x1 - x3)\,(x1 - x2)\,(x0 - x3)\,(x0 - x2)\,(x0 - x1) \tag{3}$$